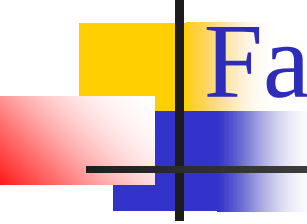


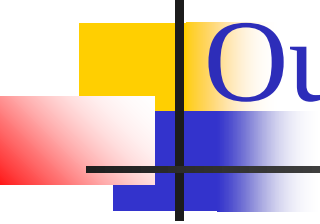
CNN TECHNIQUES

Dr. Nguyen Thi Kim Ngan



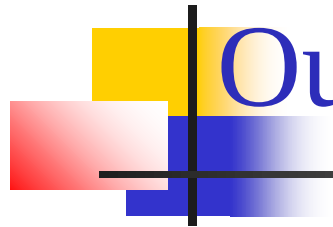
Factors contributing to poor predictive performance

- Insufficient and non-representative dataset
- Class imbalance in the training data
- Excessive complexity in model architecture
- Challenges encountered during the optimization process



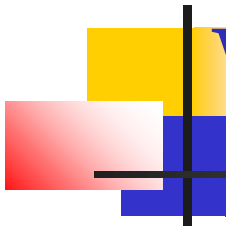
Outline

- Transfer learning
- Data Augmentation
- Data Synthesis



Outline

- **Transfer learning**
- Data Augmentation
- Data Synthesis

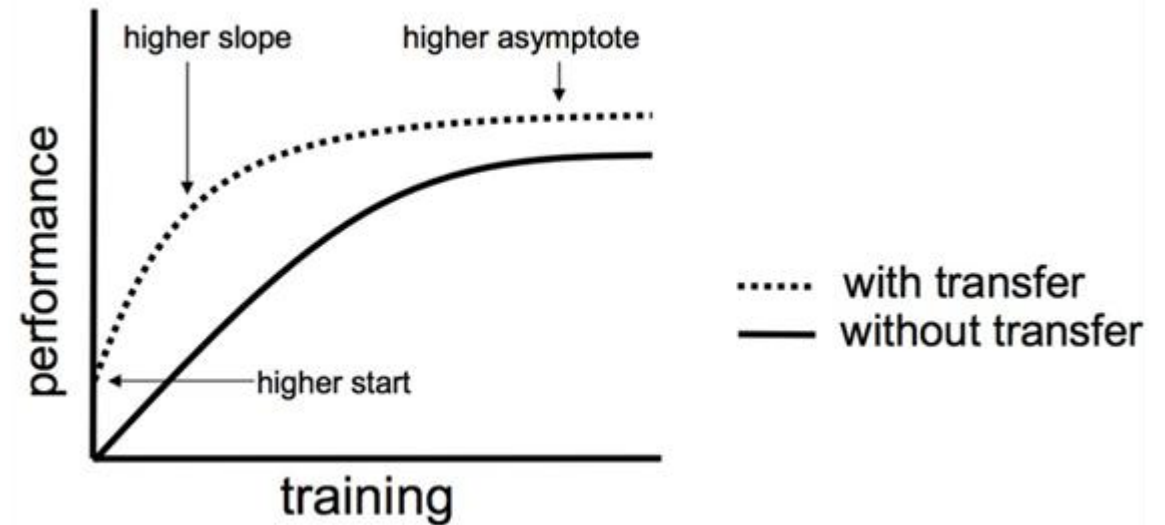


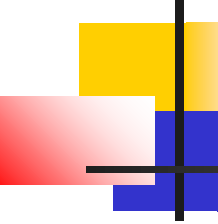
What is Transfer Learning?

- The process of applying knowledge learned from a pretrained model to the current task is referred to as transfer learning
- Transfer learning leverages the features learned from pretrained models

The significance of Transfer Learning

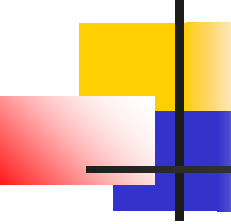
- **Knowledge transfer:** Leverages pretrained models for new tasks
- **Improved efficiency:** Fewer epochs, higher accuracy, lower training cost
- **Effective with limited data:** Enhances prediction by combining pretrained and new data





Example: Dog vs. Cat Classification

- Labels **dog** and **cat** already exist in ImageNet
- Leverage **pretrained weights** from ImageNet models
- Achieve **faster training** and **higher accuracy**



Phases of Transfer Learning

1. Pretraining phase

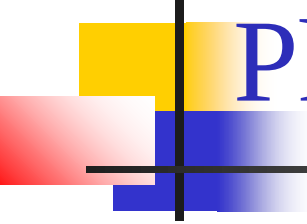
- The model is trained on a large-scale dataset (e.g., ImageNet)
- Learns general features

2. Transfer phase

- Reuse the pretrained weights
- Either keep them fixed (feature extraction) or fine-tune some/all layers

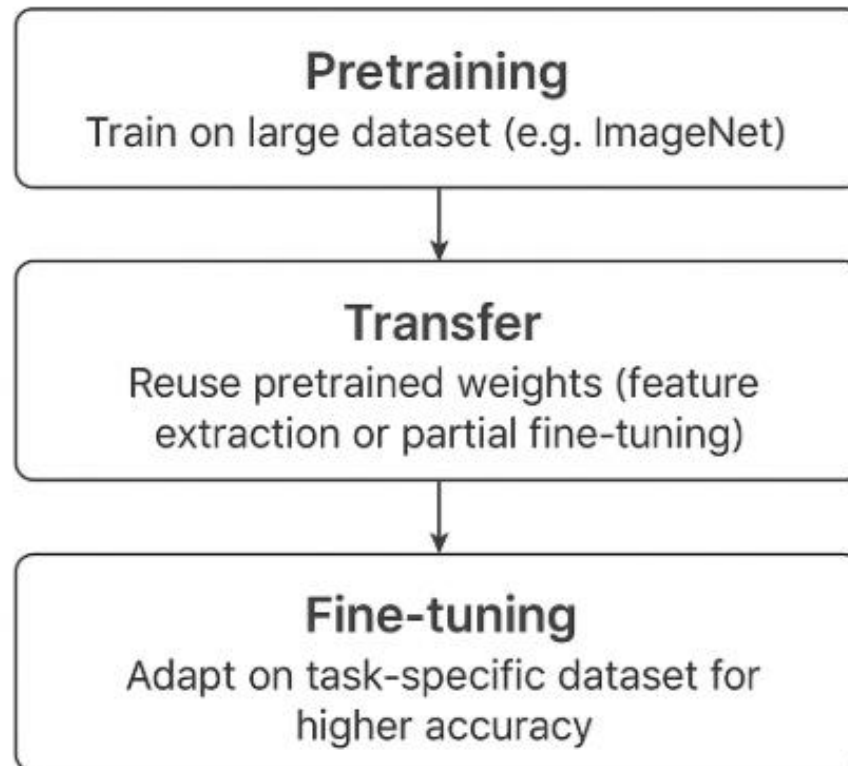
3. Fine-tuning phase (optional)

- Adjust the model on a smaller, task-specific dataset
- Improves accuracy and generalization.



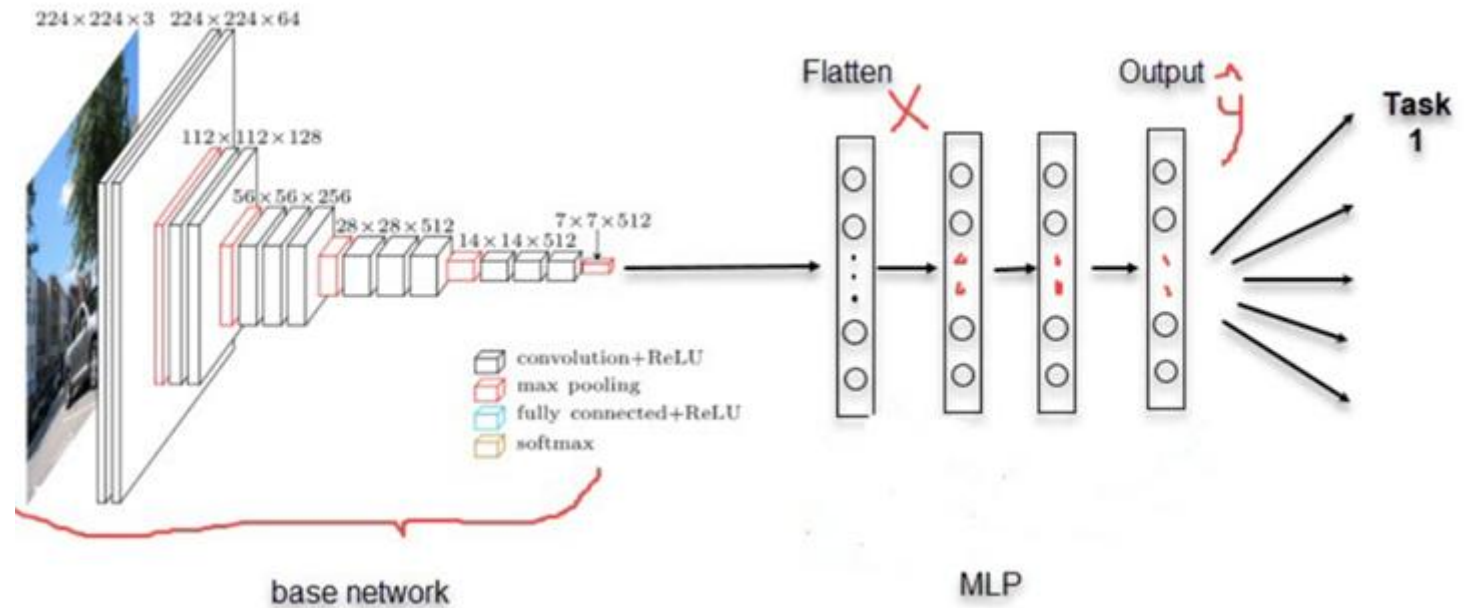
Phases of Transfer Learning

Phases of Transfer Learning



Phases of Transfer Learning

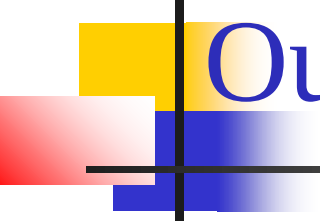
- Parameter Initialization: Base network:
 - The model reuses the pretrained weights of the base network
 - MLP network: The weights (W) and biases (b) of the MLP are initialized randomly
- After passing through the base network, the data produces meaningful features, which serve as the input to the MLP





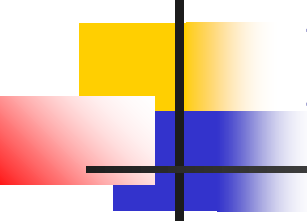
Example

```
1 from tensorflow.keras.models import load_model, Sequential
2 from tensorflow.keras.layers import Dense, Flatten
3 from tensorflow.keras.applications import MobileNet
4 from tensorflow.keras.optimizers import Adam
5
6
7 base_network = MobileNet(input_shape=(224, 224, 3), include_top = False, weights = 'i
8 flat = Flatten()
9 den = Dense(1, activation='sigmoid')
10
11 model = Sequential([base_network,
12                     flat,
13                     den])
14 model.compile(optimizer=Adam(), loss='binary_crossentropy', metrics = ['accuracy'])
15 model.summary()
```



Outline

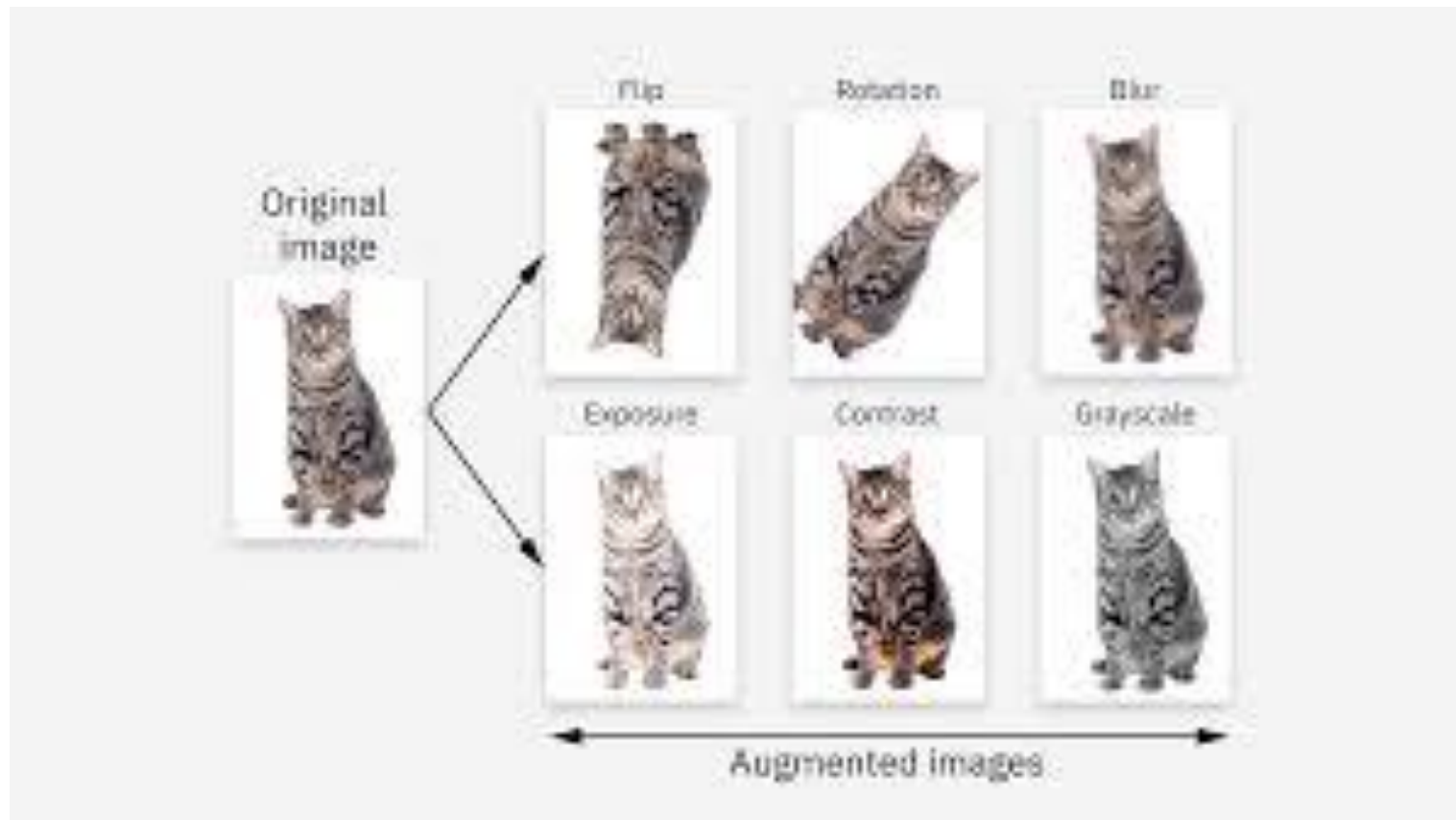
- Transfer learning
- **Data Augmentation**
- Data Synthesis

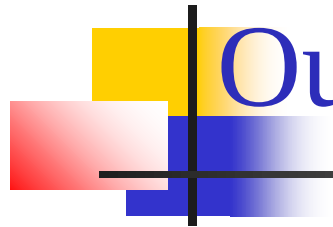


Data Augmentation

- **Definition:** Technique to artificially increase training data by applying random transformations.
- **Common Methods:**
 - Rotation, flipping, cropping
 - Scaling, shifting, zooming
 - Color adjustment, noise injection
- **Goal:** Ensure the model never sees the same image twice during training.
- **Benefits:**
 - Reduces overfitting
 - Improves generalization
 - Enhances model robustness

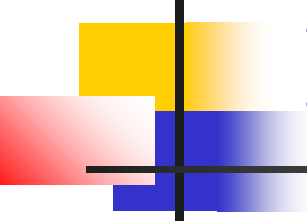
Example





Outline

- Transfer learning
- Data Augmentation
- **Data Synthesis**



Data Synthesis

- **Definition:**

Artificially generating new data samples to enrich the training set.

- **Techniques in Computer Vision:**

- Image blending and composition
- Style transfer
- GAN-based image generation
- Synthetic environments / simulations

- **Goals:**

- Overcome limited or imbalanced datasets
- Improve model robustness and generalization
- Reduce cost and effort of manual data collection

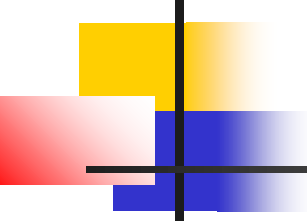


Image Loading Example

<https://www.kaggle.com/code/msripooja/dog-images-classification-using-keras-alexnet>